Sustainable Urban Transportation
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Mapping Pedestrian Fatalities Near CTA Stations and Proposing Subsequent Infrastructure Changes

Almost every CTA trip begins and ends on foot, yet many stations sit on wide, busy streets that are hard for pedestrians to cross. For our final project, we will examine which CTA rail stations have the highest concentration of crashes and pedestrian fatalities nearby, analyze what street design factors those stations share, and suggest what street design changes can be made to improve safety. To achieve this, we will draw on crash and pedestrian fatality data from the Chicago Data Portal and CTA Open Data, as well as imagery from Google Maps/Streetview in order to analyze street design. We will additionally create a framework in order to analyze each station in a consistent manner, most likely by establishing a standard radius around each station from which data will be drawn. For example, we plan to use a quarter-mile radius and count pedestrian crashes from 2019 to 2024, including crashes with serious injuries, since fatalities alone are rare. For the stations with the most crashes, we will compare features such as lane count, crosswalks and signal timing, and recommend changes such as curb extensions or leading pedestrian intervals.

Research Question: Which CTA rail stations have the highest concentration of pedestrian crashes and fatalities nearby, what street design factors do they share, and what infrastructure changes could make walking to transit safer?

This research question is highly relevant to the broader topics of environmental, social, and economic sustainability; encouraging citizens to make environmentally conscious decisions - like utilizing transit as opposed to a private vehicle - is much more difficult when certain stations are known for being dangerous for pedestrians. Furthermore, a continued lack of interest in improving pedestrian safety around problematic stations threatens the long-term economic sustainability of the CTA, as this lack of action will certainly cause at least some transit users to switch to other forms of transit, eventually leading to lower ridership. For many riders, the most likely alternative is driving, which would increase congestion and emissions. Finally, this topic is also extremely relevant to social sustainability, as we predict that many of the most dangerous stations are located in Chicago's more marginalized and economically challenged neighborhoods; in order to maintain a well-functioning, sustainable transit system into the future, the CTA must ensure equal transit safety for all Chicagoans, regardless of social or economic status.

Our final report will include a map of crash hotspots around CTA rail stations, profiles of the highest-risk stations, and design recommendations for the CTA and the Chicago Department of Transportation.

Preliminary Sources

City of Chicago. (2017). Vision Zero Chicago action plan 2017-2019. Chicago Department of Transportation.https://www.chicago.gov/content/dam/city/sites/completestreets/pdfs/17_0612%20VZ%20Action%20Plan_FOR%20WEB.pdf

City of Chicago. (n.d.). Traffic crashes - crashes [Data set]. Chicago Data Portal. https://catalog.data.gov/dataset/traffic-crashes-crashes